Warranty Claim Packet

UGREEN Hackathon 2026 on AI NAS, Round 1 proposal. Track: Automate Your Workflow. Skill name: `warranty-claim-packet`.

1. What does your Skill do?

It finds receipts, warranty terms and serial photos scattered across the NAS, shows which purchases are still covered, and assembles a ready-to-send claim packet when something breaks.

2. Who is the target user, and what is their problem?

Households and small offices that back up phones and invoices to their NAS. When a dishwasher or laptop fails, the receipt is an old phone photo, the manual a PDF in Downloads, the serial on a label photo. Unable to prove the date or check the term quickly, people pay for covered repairs.

3. What is your solution?

Ledger: one table of purchases with proof file, term, expiry and status (covered, expiring soon, expired, term unknown, no proof found). **Claim**: for one broken item, a dated verdict citing the warranty clause, a claim letter drafted only from verified facts, and a folder of copied evidence.

4. What is the main user flow?

1. User: "My dishwasher stopped draining. Am I still covered?"
2. Skill confirms item and fault, searches documents and photos.
3. Extracts purchase records, serial number and warranty clause.
4. Shows the verdict with sources and a draft letter to edit.
5. After approval, saves Warranty/Claims/[item, date] with evidence copies, letter and checklist.

5. How is your idea different from existing tools or common AI solutions?

Chat assistants answer from what you paste; file search matches names. This joins invoice PDFs, receipt photos and manuals into one cited verdict, never guesses a date or term ("term unknown" is a status), and outputs a packet support agents accept. Financial documents never leave the NAS.

6. Which UGREEN AI NAS features does your Skill use, and at which step?

Step 1: file-workflow get-user-dir. 2: ug-filesystem find. 3: ug-photo OCR search. 4: ug-document extract (receipt, invoice schemas). 5: ug-photo vision-ocr, vision-vlm on the local model. 6: ug-document qa. 8: ug-document writing. 9: ug-filesystem mkdir, cpmv (copy), ug-knowledge save-text, ug-filemgr tags.

7. What is your technical approach?

Pure orchestration SKILL.md (Uliya standard) delegating to file-agent, photo-agent, document-agent and knowledge-agent. Named values carry real expert results between steps; a missing value stops dependent steps. Read-only until the user approves a write; originals are copied, never moved. All five delegation states handled; card numbers masked. Inputs: mode, item, fault, optional scope.

8. What have you built so far? What is not built yet? What still needs testing on a NAS?

Built: complete SKILL.md, coverage rules with date edge cases, packet spec, disclosure; official checker 0.7.3 reports static-pass, 0 errors, 0 warnings. Not built: keyword tuning on real libraries. Needs a NAS: OCR recall on receipt photos, extraction from order-confirmation PDFs, approval prompts for writes, tag creation.

9. How will you test your Skill if you become a Finalist?

Core 1: ledger over 8 seeded purchases (PDF invoices, receipt photos, 4 manuals); each row needs correct expiry, status and sources. Core 2: claim for a dishwasher bought 14 months ago with a 24-month term; expect "covered", a letter, 3 copies and 2 Markdown files, originals unchanged. Edge: illegible receipt date, no manual; expect "no proof found", no letter, nothing written.

10. What are the known limits, risks, or outside dependencies of your Skill?

Depends on OCR quality and on manuals stating a term. Credit-card extensions and statutory guarantees are not evaluated unless a stored document states them. Runs cap at 300 documents and 60 photos. No outside dependencies.

Entrant: Matthew Busel (individual entrant). Electronic signature: /s/ Matthew Busel, November 1, 2026.

Third-party components and licenses: none bundled; the package contains only Markdown written by the entrant. Built with the official UGREEN Skill Developer Package (uliya-skill-kit 0.7.3, capability catalog 2.0.0) under the UGREEN Skill Hub User Agreement and publishing guidelines; no kit file is redistributed. AI tool: Claude (Anthropic) in Claude Code for drafting and checking, directed and reviewed by the entrant. Source: github.com/Mattbusel/warranty-claim-packet (MIT).